

Integrated Sensing and Communication in 6G: Motivations, Use Cases, Requirements, Challenges and Future Directions

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Abstract— In 6G, it is envisaged that the function of sensing and communication will coexist and be fully integrated in one system, sharing the same resources in time, frequency and space domains, as well as key elements including waveform, signal processing, hardware, etc. This paper discusses novel applications, key performance requirements, challenges and future research directions for Integrated Sensing and Communication (ISAC) design in 6G. First, four categories of ISAC use cases are described as new services in 6G together with their corresponding performance requirements on 6G design. In addition, it is demonstrated how the information obtained through sensing can significantly improve the performance of communication. Thereafter, the challenges in the design and evaluation of a practical ISAC system are discussed, including sensing and communication performance tradeoffs, hardware imperfections, as well as the investigation of an appropriate channel model. Finally, future research directions are presented to set the path for designing an efficient ISAC network.

Keywords—*Integrated Sensing and Communication (ISAC), 6G, sensing assisted communication, localization, imaging, simultaneous localization and mapping (SLAM), augmented human senses, gesture and activity recognition*

I. INTRODUCTION

Cellular mobile communication systems have evolved dramatically since its inception in 1980s. New generations of cellular networks emerge every 10 years or so and it took almost four generations to have everyone connected anywhere and anytime, leading to a connected society. With the rapid world-wide commercialization of 5G, not only will societies be better connected with enhanced communication capability, such as faster speed and lower latency, but also more devices such as cars, buildings, and even appliances will be connected. We are moving from a connected society to connected everything, which will eventually enable every business to exploit the next wave of digitalization. We envision that 6G will continue to transform from connected people and connected things, to connected intelligence. The mobile network will connect a large amount of intelligent devices in a smart fashion. These intelligent nodes will have the capability to sense their surroundings, and exchange their observations through communication. In addition, Artificial Intelligence (AI) can be combined with sensing capability, for the nodes to fuse the physical and biological worlds with cyber worlds such that the network will have human like cognition capabilities.

The use of higher frequency bands (from mmWave up to THz), wider bandwidth, and denser distribution of massive antenna arrays in future 6G systems will enable the integration of wireless signal sensing and communication in a single system to mutually enhance each other. On one hand, the communications system as a whole can serve as a sensor. It can explore the radio wave transmission, reflection, and

scattering to sense and better understand the physical world, providing a broad range of new services. On the other hand, the capabilities of high-accuracy localization, imaging, and environment reconstruction obtained from sensing could help improve the communication performance such as more accurate beamforming, faster beam failure recovery, and less overhead to track the channel state information (CSI).

The integration of sensing and communication functions can happen at different levels, from loosely coupled to fully integrated, from shared spectrum, shared hardware, to shared signal processing and protocol stacks, and even cross-module, cross-layer information sharing, benefiting one another. With ISAC, it is easy to imagine that an autonomous driving car could work well or even better in all types of weather and light conditions without being equipped with separate and high cost radar and lidar.

For the rest of the paper, we will discuss about the potential use cases of ISAC and the performance requirements therein. We begin with a general overview of four categories of use cases and new performance metrics brought by sensing to the communication system in section II. Then, we dive into each of the use cases to analyze the detailed performance metrics in section III. After that, we discuss about the potential challenges in the design of an ISAC system as well as the future research directions in section IV. Finally, section V concludes the paper.

II. OVERVIEW OF ISAC USE CASES AND NEW PERFORMANCE INDICATORS

6G will feature the networked sensing capability. The ISAC system will provide high-resolution sensing, localization, imaging, and environment reconstruction capabilities to improve communication performance and at the same time open up a broader range of network service scenarios, laying a data foundation for building an intelligent digital world. To avoid plainly listing all potential new services that could be supported by future ISAC systems, Fig. 1 summarizes them into four categories, namely

- 1) High-accuracy localization and tracking;
- 2) Simultaneous imaging, mapping, and localization;
- 3) Augmented human sense;
- 4) Gesture and activity recognition.

For instance, the ISAC system can provide both device-based and device-free localization services for 6G, which leads to supporting use cases requiring high-accuracy localization such as automatic docking for drones and collaboration between robots for complex tasks. In addition, 6G based sensing capabilities in simultaneous imaging, mapping, and localization enable the mutual performance improvements of these functions which opens up the realm of possibilities in 3D indoor/outdoor non-line-of-sight imaging and mapping.

On the other hand, with the use of much higher frequency bands, the ISAC system can be implemented in portable devices to augment human senses and enable people to “see” beyond the limits of human eyes. This may sound a bit sci-fi but ISAC will be very helpful in future smart hospitals to obtain information on vascular/organ status and other vital signs. Device-free gesture and activity recognition based on the joint capability of sensing and machine learning is another promising aspect of ISAC applications to promoting contactless user interfaces and camera-free supervision of the patients or elderly people so that emergencies can be identified in time while privacy can be protected. The detailed descriptions of these new services and the corresponding performance requirements in each category will be described in the next Section. Moreover, it is worth to mention that, in addition to sensing as a new service, sensing can also be used to assist communications and positioning, more details of which can be found in Section III.E.

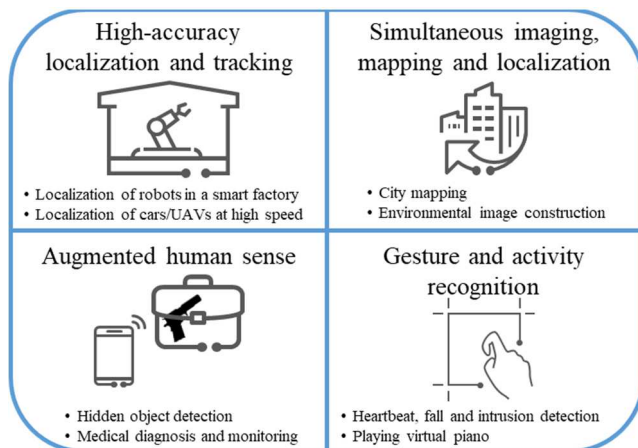


Fig. 1. Four categories of ISAC use cases as the new services in 6G.

Within the ISAC context, several new key performance indicators (KPIs) are introduced for sensing capability such as: 1) Sensing accuracy which is the difference between the sensed and real values in range, angle, velocity, etc., 2) Sensing resolution which is the separation between multiple objects in range, angle, velocity, etc. and 3) Detection/False alarm probabilities which is the probabilities that an object will be detected when one is present/not present.

III. SENSING APPLICATIONS AND KEY PERFORMANCE REQUIREMENTS

A. High-accuracy Localization and Tracking

Wireless infrastructures provide millions of terrestrial anchors which enables tens of meters level of accurate localization for mobile network terminals, no matter indoor or outdoor, in clear or bad weather, stationary or moving. Typical positioning techniques include time of arrival or angle of arrival based location estimation (e.g. by using trilateration and triangulation methods). On top of this, 4G and 5G evolves with higher bandwidth and larger antenna number, which enables fine CSI fingerprint acquisition and this brings about meter level accuracy for vehicles on road and robots/IoT in factories [1]. The highest KPIs (supported by 5G NR Rel-17) are given as 1 m horizontally, 3 m vertically, 10 ms latency in commercial scenarios and 0.2 m horizontally, 1 m vertically, 10 ms latency in Industrial IoT scenarios [2]. However, it is still not accurate enough to support services such as in future intelligent factories or those involving collaborative robots. For instance, the capabilities

of future assistant robots will approach that of human in daily work and life. To adapt to human’s living and working environment, high precision in centimeter level (e.g. 2 cm) is needed such as to find and rotate door handle, pick up the right box/book from the shelf, and capture the right tool from the drawer. With ISAC capability in 6G, such accuracy could be achievable with the same system for wireless communication, which is very useful in the case of cost and privacy preservation, or in extremely harsh conditions where camera or lidar could not work.

High precision localization is also very much desired for autonomous systems that are becoming pervasive in smart manufacturing when two or more communicating devices operate in collaboration. For instance, a drone may need to land on a moving carrier vehicle to get charged; a delivery robot may need to refill liquid or solid substance to a smart container (bin/tank) when it is detected as empty, etc. In such proximity use cases, sub-centimeter level (e.g. 5 mm) relative localization accuracy is required to perform the task. Note that due to the communication requirements and cost limitations, the radio systems among these parties can be asymmetric, e.g. simple radio frequency identification (RFID) tag can be attached to containers while more sophisticated RF arrays could be equipped on delivery robots.

In the future, the localization task may not only be fed with target coordinates as its destination, but also some semantic destinations that would need AI or communication technologies to interpret from some learnt linkage between the physical coordinates and task specific destinations. The destinations can be locations within reachable area, which has the best channel condition to get the messages transmitted, or the real time location nearby the robot’s master, e.g. a coffee table next to the human host, as a perfect place to place a cup of hot tea. In such cases, the ISAC system needs to learn the location and understand the semantic meaning. A human like common sense of geometric environment is important for the future wireless localization system.

B. Simultaneous Imaging, Mapping and Localization

Besides the high-accuracy localization, joint mapping and localization can enable robots or machines to be trained to have human-like senses. For instance, a robot can obtain the human’s current location with respect to the locations of the furniture in the room (mapping) by device-free sensing. Then based on the sensing results and with the support of AI, it can derive the route and deliver a target to the human. The technique described above is commonly known as simultaneous localization and mapping (SLAM). The concept of SLAM is to use the channel multipath information to reconstruct 2D images and 3D models of the environment, which at the same time help provide more reliable and accurate positioning service. From another aspect, legacy localization schemes require physical transceivers (PT) as anchors. Their performance is bounded due to the limited number of such anchors. Environment sensing can allow devices to identify virtual transmitters (VT, refers to mirrored image of physical transmitter reflected by the surrounding walls) such as reliable usable fixed landmarks as virtual anchors and thus improve the localization accuracy.

Fig. 2 shows that with SLAM, accurately learnt VT locations can help achieve UE localization accuracy of 10 cm, while conventional localization based on the delay estimation from raw multi-path components can only offer 60 cm accuracy (60%). 6-fold improvement is thus demonstrated

from the test shown in the Figure. Meanwhile, the estimation of VT locations can be accumulated over time and thus gets refined incrementally. Moreover, the location of walls can be derived from the continuous presence of VTs during the SLAM procedure. It is worth noticing that due to higher moving speed, the mapping accuracy by autonomous vehicles in outdoor scenarios is usually lower than those in indoor cases with low mobility devices. While the mapping accuracy for some dedicated purpose of 3D building/environment reconstruction can achieve 1 cm accuracy when the mapping device moves with low speed and stable orientation.

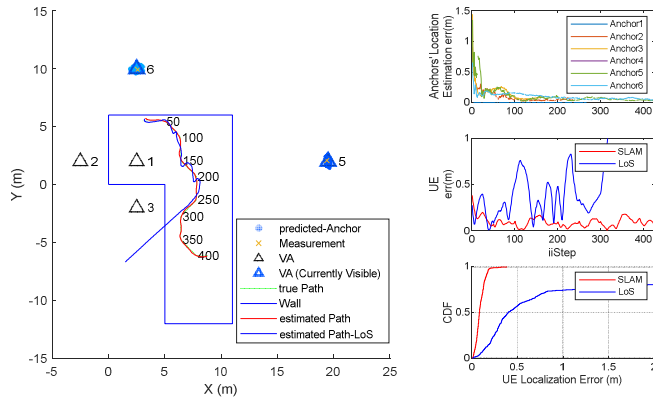


Fig. 2. SLAM provides accurate localization for both UE and VTs. The estimation of VTs' location converges with time. With the VT location learnt from the NLOS paths, UE localization error is much smaller than that of the LOS-path only cases.

C. Augmented Human Sense

RF sensing techniques support high-resolution imaging and detection with coherent signal accumulation. With 6G networks, these will open the door for numerous applications, such as remote surgery, detection of slits on products, and sink leakage detection. For this type of application, very high range and cross-range resolutions are required, which rely on tens GHz bandwidth and thousands antenna elements, respectively.

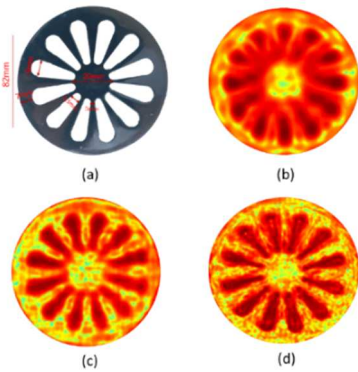


Fig. 3. The imaging result of a lemon core for different frequencies: a) the target's optical image; b) imaging results at 73 GHz; c) imaging results at 140 GHz; d) imaging results at 220 GHz.

An example of millimeter-level imaging resolution is shown in Fig. 3. While Fig. 3a is an optical photograph of the target (lemon core) to be imaged, Fig. 3b, 3c and 3d are the wireless imaging results at 73 GHz, 140 GHz, and 220 GHz, respectively. A virtual array scanning mechanism is applied with half wavelength spacing, forming an equivalent aperture of about 20 cm $\times$ 20 cm. The resolution improves from 8 mm

to 4 mm and to 1 mm, as the bandwidth and frequency increases. Different from optical-dependent human vision, RF sensing could be considered as an all-day solution thanks to the propagation characteristic of radio waves. The sensing devices can be 6G-enabled mobile phones, wearables, or medical equipment implanted beneath the human skin. With the help of scientific and technological advancement, we can achieve augmented human sensing to facilitate information collection in darkness, inside a box, and beneath the skin.

6G sensing can also play an important role in the medical technologies applied to diagnosis, monitoring, and treatments. Chronic diseases (asthma, arrhythmia, hypoglycemia, etc.) need to be constantly monitored in the long term using wearable devices. Sensing therefore helps doctors instantly acquire accurate information on a patient's condition such as heartbeats, detecting real-time images of the blood stream beneath the skin. In an environment with low visibility (i.e. at night, fog, or glare), the wireless imaging technology can be conveniently applied in enhanced sensing glasses and vehicular devices, to greatly facilitate pedestrian and vehicle in a transportation system.

In addition to the spatial detection and imaging, the widespread use of higher frequencies will enable spectrogram application. Spectrogram recognition is based on identifying targets through the spectrogram sensing of their electromagnetic or photonic characteristics such as absorption, reflectivity, and permittivity parameters, which helps to distinguish the materials. It can be captured under the high frequencies such as THz and NIR (Near infrared). For instance, the PM2.5 transmission spectrum has two distinct absorption bands between the 2.5 and 7.5 THz [3]. It indicates that the metallic oxides' absorption falls within the range of 2.5 – 7.5 THz. These results verified that the spectral analysis is a promising tool, which can be utilized to understand the composition of materials.

D. Gesture and Activity Recognition

The higher frequency band in 6G will enable higher resolution and accuracy in order to capture finer activities and gestures. The ubiquitous artificial intelligent techniques with amazing computing power also open up a new era for activity and gesture recognition. In the foreseeable future intelligent device-free activity recognition and gesture capturing system, sensing will not be limited to a single home but rather in a large-scale complex indoor environment. Other than the UE (e.g. smart phone), the indoor cellular transmission point (TP) or base station (BS) will also be used as a sensor. They will be jointly used to sense the surroundings and thus the fused sensing performance would be greatly improved. In addition, the indoor cellular TP/BS allows a remarkably extended sensing range that is not reachable through UEs. These fused sensing information are also communicated with cloud-based services or among adjacent buildings that are enabled by the cellular TP/BS.

With high classification accuracy, many functionalities such as gesture recognition, heartbeat detection, fall detection, respiration detection, sneeze sensing, intrusion detection, etc. can be implemented in a smart hospital in the foreseeable future. As a novel use case, the medical rehabilitation system in the smart hospital enables automatic supervision of patients during their physiotherapy exercises. There will be automatic prompt alerts on incorrect movements or gestures, thus significantly improving the patients' rehabilitation. In terms of some advanced

contactless functionalities such as playing virtual piano, more precise capturing capability and ultra-high classification accuracy will be required. Considering the average width of the black keys is about 10 – 11 mm, to locate the finger height above the board and the horizontal position relative to the virtual piano keys, the accuracy is anticipated to be about 3 mm or better. In addition, the probability of recognition should be above 99% to ensure a smooth melody.

E. Sensing Assisted Communication and Localization

Although sensing will be introduced as a separate service in the future, it might still be beneficial to look at how the information obtained through sensing can be used in communication. The most trivial benefit of sensing will be environment characterization, which enables medium-aware communication due to more deterministic and predictable propagation channels. It has been shown that the environment knowledge provided by sensing not only improves the accuracy of channel estimation in mmWave, but it also significantly reduces the overhead since the environment is shared by potentially many UEs and sensing-based channel acquisition avoids repeating the channel estimation process for each individual link [4]. Another example would be sensing-assisted beam alignment, especially in mmWave vehicular communication where the main challenge is the frequent link reconfiguration resulting in significant overhead. In [5], it has been proposed to use the information obtained from a radar mounted on the infrastructure operating in a given mmWave band to configure the beams of the vehicular communication system operating in another mmWave band. Moreover, the users' location information and the environment map obtained by sensing helps identifying the link blockage caused by large objects, especially in dense urban networks, and improve the communication throughput [6]. Other examples of sensing-assisted communication can also be considered and studied to reduce the latency and overhead and improve the performance of communication systems in the future 6G networks by the help of information provided by the sensing system. Another benefit of sensing for communication would be improving the users' positioning accuracy through combining the advantages of the active localization and passive localization and overcoming their shortcomings to satisfy the 6G localization requirements.

IV. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

A. Fundamental Tradeoffs

Let us consider a scenario where an RF source S is transmitting a signal $x(t)$, which is propagated through the communication medium $\mathcal{M}$ to yield $Y(t)$, the received signal at the intended receiver. In communication, the RF signal $x(t)$ transmitted by the RF source is manipulated by the information available at this source (d_S), whereas in sensing, the RF signal is manipulated by the information available at the communication medium (d_M). In other words, sensing can be interpreted as a communication method to convey the information embedded in the communication medium by transmitting an RF signal from a different node. This suggests to define ISAC as a general framework in which any RF signal can be used to convey both the communication data (d_S) and the sensing data (d_M). Since both data are communicated using a single RF source, a fundamental tradeoff seems to exist between the communication and

sensing performance and characterizing this tradeoff helps better understand the design criteria for an ISAC network.

One well-known example of ISAC in the current communication systems is the channel estimation process in which the pilot/reference signal (sensing signal) is transmitted along with the data-carrying signal (communication signal) and $d_M = H$, the propagation channel. It can be seen that a fundamental tradeoff appears here; the more pilot resources used, the better channel estimation performance is obtained at the expense of communication spectral efficiency. Note that in this specific example, sensing (channel estimation) is only used to assist the communication, i.e. the sensing data (propagation channel) is only used to help the receiver decoding the data embedded in the communication signal. In a general ISAC framework with sensing introduced as a separate service, the sensing information will be valuable on its own.

One of the first attempts in characterizing the tradeoff between the communication and sensing performance in a more general setup is reported in [7], which considers a single-antenna receiver capable of processing the target echo signal and the UL communication signal simultaneously. The key idea to characterize the tradeoff is to see this setup as a multi-access (MAC) channel considering the target as a virtual communication user. Defining the “radar estimation rate” as the mutual information between the radar and the target, the communication and sensing performance tradeoff is characterized as the capacity region of the communication data rate versus the radar estimation rate under different multi-access strategies [7]. Another notable contribution in formulating the tradeoff between communication and sensing performance is [8], which proposes a single-objective ISAC system. As shown in Fig. 4., the channel input, output, feedback output, and the state random variables of the environment target (also called “state”) are denoted by X_i , Y_i , Z_i , and S_i , respectively, and the joint probability distribution of the proposed model is given by

$$P(\omega, x, s, y, z) = P(\omega) \prod_{i=1}^n P(s_i) \prod_{i=1}^n P(x_i | \omega z^{i-1}) P(y_i z_i | x_i s_i). \quad (1)$$

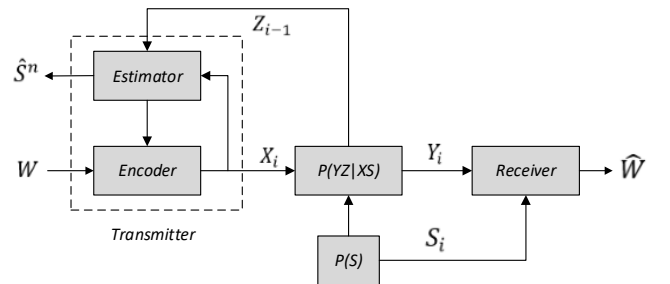


Fig. 4. ISAC in a single-objective system [8].

For this ISAC framework, the communication performance is characterized by the achievable rate R and the sensing performance is characterized by the distortion measure. A rate distortion pair (R, D) is said to be achievable if there exist $(2^{nR}, n)$ codes with

$$\lim_{n \rightarrow \infty} P(\hat{W} \neq W) = 0 \quad (2)$$

and

$$\lim_{n \rightarrow \infty} \sup \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n d(S_i, \hat{S}_i) \right] \leq D \quad (3)$$

The capacity-distortion tradeoff is defined as the supremum of R such that (R, D) is achievable. For the optimization target of system design in Fig. 4, the transmitter of the sensing sub-system expects that the echo wave Z_{i-1} is better when it is stronger, and the transmitter of the communication sub-system expects that the signal Y_i captured by the receiver is also better when it is stronger. However, the total energy of the transmitted signal X_i is fixed, which results in a tradeoff between the sensing performance and communication performance. It would be desired to characterize this tradeoff in more general setups to get better understanding for designing ISAC systems based on the given KPIs in various scenarios.

B. Hardware Imperfections

In the design of the ISAC system, baseband and RF hardware sharing is an important area of focus. The hardware integration solution reduces the overall power consumption, system size, and information exchange latency between the two systems. Meanwhile, the hardware converging strategy would facilitate the mutual aided functions of sensing and communication in distortion calibration and compensation. The common impairments in ISAC system due to hardware imperfections are demonstrated in Fig. 5.

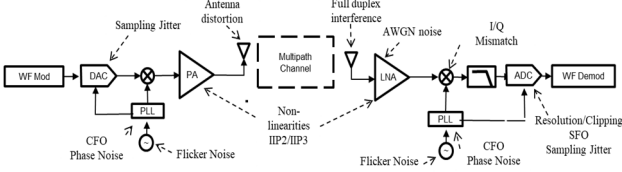


Fig. 5. Impairments of ISAC transmission system.

It should be noted that, in light of differences in evaluation metrics and algorithms between communication and sensing, hardware requirements are quite different. Considering the cost and size of the historical communication and radar systems, the ISAC system hardware design would be closer to the traditional communication architecture. As a tradeoff, we need to consider the impact of distortion parameters on sensing performance. For instance, a communications system hinges on full duplex isolation to achieve high capacity. In contrast to that, from the OFDM ISAC perspective, limited transmitter-receiver isolation is a primary concern in the detection of static targets [9]. Proper design of integrated RF architecture and self-interference cancellation in an ISAC system are key technical problems to solve. Another issue is that sensing requires the accumulation of coherent signals to ensure performance, which makes the system more sensitive to sampling jitter, frequency offset and phase noise [10]. This in turn leads to high requirements on synchronization and stability of the system. In short, we need to consider these hardware challenges in the selection of ISAC waveforms, sensing algorithms and impairment compensation schemes.

C. Channel Modeling

The algorithm design and performance evaluation of sensing use cases strongly depend on the location of the target and surrounding environment, which cannot be realized by stochastic channel models. We therefore anticipate the adaptation of deterministic channel modeling methodologies that may lead to more precise evaluation. Channel models

may be typically classified into deterministic, stochastic and quasi-deterministic as depicted in Fig. 6. Deterministic models include computational electromagnetics (CEM), ray tracing, and measurement models, which are related to more deterministic scenarios. In addition, stochastic models which include geometry-based stochastic model (GBSM), and non-geometry stochastic model (NGSM) [11] are also considered as strong candidates for communication evaluation due to their brevity in large-scale simulation.

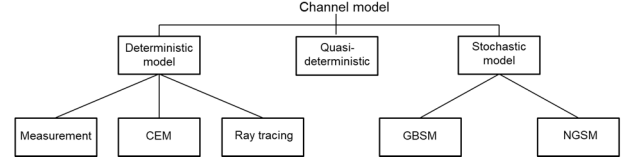


Fig. 6. Typical channel model classification.

For sensing uses cases such as localization and mapping, ray tracing could be considered as a strong candidate for channel modeling methodology. However, as another typical application of sensing, imaging needs to take CEM into account when the size of an object is approximately close to a wavelength, since it is difficult to model this through the geometric optics method. Based on the preceding discussion, we can see that a single scheme may not meet the evaluation requirements of ISAC applications. Research on the channel model historically tended to focus on improving the deterministic models under limited complexity. This might therefore be the potential direction of an evolving hybrid model, in which a plurality of mechanisms is included, but the weights of the mechanisms vary according to different application scenarios and evaluation criteria.

D. Joint Waveform Design and Optimization

Most works on the joint design of sensing and communications focus mainly on joint waveform design. The main challenge for the joint waveform design is the contradicting KPIs for communications and sensing. In particular, the main target for communications is maximizing the spectral efficiency, while the optimum waveform design for sensing targets estimation resolution and accuracy. Since CP-OFDM has been proven to be a favorable option for communication, many researchers have considered this waveform for sensing as well. Although the introduction of cyclic shift (CP) has been shown to degrade auto-correlation in the time domain [12], a novel approach of frequency domain processing [13] allows for efficient parameter estimation of CP-OFDM, achieving the maximum processing gain. Furthermore, CP-OFDM has been shown not to have range-Doppler coupling problem, which means that the range and Doppler estimation can be performed independently [13]. However, these favorable properties of CP-OFDM counts on perfect synchronization (in both time and frequency domain) assumption between the transmitter and the receiver which may not be present, especially for bistatic sensing. Not to mention that large peak to average power ratio of CP-OFDM is another major issue for radar applications where power efficiency is very important.

Alternatively, frequency modulated continuous wave (FMCW) waveform, which has traditionally been used for radar, is not capable of carrying data at transmission rates desirable for communication services. Some researchers proposed to modify the FMCW waveform to make it more communication-friendly. Among the many contributions in this line of research, we can mention [14], in which the

authors propose to use up-chirp for communication and down-chirp for radar and [15], introducing trapezoidal frequency modulation continuous-wave (TFMCW) modulation in which the radar cycle and communication cycles are multiplexed in time domain. Although these techniques enable efficient multiplexing of communication data in the sensing signal, they still suffer from the low spectral efficiency due to the existence of the chirp-like sensing signal. Another line of research is devoted to using single-carrier waveform based on code domain spreading of joint radar and communication signal [16]. For this class of waveforms, the radar performance has been shown to be affected by the auto-correlation of the sequences and the long spreading codes result in good auto-correlation at the expense of communication spectral efficiency [15]. In addition, Doppler estimation requires more complicated algorithms [15]. Current state of the art suggests that there is still room for waveform design to strike a balance between good communication and sensing performance.

E. Artificial Intelligence Enabled ISAC

AI is regarded to be an efficient method of dealing with intractable problems that we faced in ISAC. For instance, learning-based approaches could be used to improve the sensing service such as gesture recognition [17] and imaging [18]. Moreover, in cooperative ISAC scenarios, node beliefs can be shared in spatial and temporal dimensions, thereby improving sensing performance through multi-node spatio-temporal cooperation. The RF sensing performance also depends on the data processing and fusion algorithms used in the state updates. Therefore, belief propagation algorithms and fusion algorithms with learning-based approaches can inherently benefit cooperative sensing and should therefore be studied. Owing to the acceleration of parallel computation, the AI schemes can operate faster and are thereby more applicable in practice in 6G. With these attractive characteristics, AI will open up a broad realm of possibilities in the ISAC system design.

V. CONCLUSION

From the discussion of the paper, it can be seen that ISAC is a promising direction to be investigated towards 6G. The sensing function integrated in the future communication system would open up brand new services and make 6G beyond just communication. In particular, this paper discussed applications and their key performance requirements for four categories of ISAC use cases as new services that could be offered by 6G. From the discussion, it is justified that sub-centimetre level of high-accuracy localization would be important for applications involving collaborative robots, while millimetre level of imaging resolution is necessary for numerous applications, such as remote surgery and sink leakage detection. It is worth mentioning that not only the communication signals can be reused for sensing, but also the communication network as a whole could be used to enhance the sensing performance. On the other hand, this paper also discussed how the sensing capability and the information obtained through sensing can be exploited to improve the quality of communications. Moreover, it is pointed out that there are major challenges in the practical implementations for the design and evaluation of ISAC systems, including the intrinsic difference in the design metrics and thus potential tradeoffs in sensing and communication performance,

hardware imperfections, as well as the adaptation of an appropriate channel model for their evaluation. Finally, the paper provided future research directions that would set the path for designing an efficient ISAC network in the near future. In summary, ISAC is established as one of the most disruptive technologies in 6G, upon which numerous real-time machine learning and AI applications can be created.

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